

HDOC – Experiment 25

Question

Write a C program that displays the following menu and uses a switch statement to print the “Decreasing patterns” selected by the user. Use nested loops without arrays. Each pattern may be implemented in a separate parameter less function with no return value.

1. Problem Analysis

We need to create a menu-driven C program that prints one of four decreasing patterns.

The program must:

- Display a menu with 4 choices.
- Use a switch statement.
- Use nested loops to generate the patterns.
- Not use arrays.
- Implement each pattern in a separate function.
- Each function must:
 - have no parameters
 - have void return type
- Accept the number of rows n inside the selected function.

Menu

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern

4. Reverse-number triangle

2. Algorithm

Main Program Algorithm

1. Start.

2. Declare an integer variable choice.

3. Display the menu:

-

1. Decreasing right-angled triangle

-

2. Decreasing natural-number pattern

-

3. Decreasing repeated-number pattern

-

4. Reverse-number triangle

4. Accept choice from the user.

5. Use a switch statement:

- If choice = 1, call decreasingTriangle().

- If choice = 2, call decreasingNatural().

- If choice = 3, call decreasingRepeated().

- If choice = 4, call reverseTriangle().

- Otherwise, display "Invalid choice!".

6. Stop.

Algorithm for Pattern 1: Decreasing Right-Angled Triangle

1. Start.
2. Input the number of rows n .
3. Check whether n is positive.
4. If $n \leq 0$, display "Invalid number of rows!" and return.
5. Set $i = n$.
6. Repeat while $i \geq 1$:
 - Set $j = 1$.
 - Repeat while $j \leq i$:
 - Print *.
 - Increment j .
 - Move to the next line.
 - Decrement i .

7. Stop.

For $n = 4$:

* * * *

* * *

* *

*

Algorithm for Pattern 2: Decreasing Natural-Number Pattern

1. Start.
2. Input the number of rows n .
3. If $n \leq 0$, display an error message and return.
4. Set $i = n$.
5. Repeat while $i \geq 1$:

- Set $j = 1$.
- Repeat while $j \leq i$:
 - Print j .
 - Increment j .
- Move to the next line.
- Decrement i .

6. Stop.

For $n = 4$:

1 2 3 4

1 2 3

1 2

1

Algorithm for Pattern 3: Decreasing Repeated-Number Pattern

1. Start.
2. Input the number of rows n .
3. If $n \leq 0$, display an error message and return.
4. Set $i = n$.
5. Repeat while $i \geq 1$:
 - Set $j = 1$.
 - Repeat while $j \leq i$:
 - Print i .
 - Increment j .
 - Move to the next line.
 - Decrement i .

6. Stop.

For $n = 4$:

4 4 4 4

3 3 3

2 2

1

Algorithm for Pattern 4: Reverse-Number Triangle

- 1. Start.**
- 2. Input the number of rows n.**
- 3. If $n \leq 0$, display an error message and return.**
- 4. Set $i = n$.**
- 5. Repeat while $i \geq 1$:**
 - Set $j = i$.**
 - Repeat while $j \geq 1$:**
 - Print j.**
 - Decrement j.**
 - Move to the next line.**
 - Decrement i.**

6. Stop.

For $n = 4$:

4 3 2 1

3 2 1

2 1

1

2. Test-Case Table

Test Case	Menu Choice	Number of Rows n	Expected Output	Actual Output	Result
1	1	4	Decreasing * triangle with rows 4, 3, 2, 1	Same as expected	Pass
2	2	4	1 2 3 4, 1 2 3, 1 2, 1	Same as expected	Pass
3	3	4	4 4 4 4, 3 3 3, 2 2, 1	Same as expected	Pass
4	4	4	4 3 2 1, 3 2 1, 2 1, 1	Same as expected	Pass
5	1	1	*	Same as expected	Pass
6	2	0	Invalid number of rows!	Same as expected	Pass
7	3	-2	Invalid number of rows!	Same as expected	Pass
8	5	—	Invalid choice!	Same as expected	Pass

3. Complete C Program

```
#include <stdio.h>

/* Function prototypes */
void decreasingTriangle(void);
void decreasingNatural(void);
void decreasingRepeated(void);
void reverseTriangle(void);

int main(void)
{
    int choice;

    printf("DECREASING PATTERNS\n");
    printf("1. Decreasing right-angled triangle\n");
    printf("2. Decreasing natural-number pattern\n");
    printf("3. Decreasing repeated-number pattern\n");
    printf("4. Reverse-number triangle\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            decreasingTriangle();
            break;
```

```
case 2:
    decreasingNatural();
    break;

case 3:
    decreasingRepeated();
    break;

case 4:
    reverseTriangle();
    break;

default:
    printf("Invalid choice!\n");
}

return 0;
}

/* Pattern 1: Decreasing right-angled triangle */
void decreasingTriangle(void)
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);
```

```

if (n <= 0)
{
    printf("Invalid number of rows!\n");
    return;
}

for (i = n; i >= 1; i--)
{
    for (j = 1; j <= i; j++)
    {
        printf("* ");
    }
    printf("\n");
}
}

/* Pattern 2: Decreasing natural-number pattern */
void decreasingNatural(void)
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    if (n <= 0)
    {

```

```

        printf("Invalid number of rows!\n");
        return;
    }

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

/* Pattern 3: Decreasing repeated-number pattern */
void decreasingRepeated(void)
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid number of rows!\n");
        return;
    }
}

```

```
for (i = n; i >= 1; i--)
{
    for (j = 1; j <= i; j++)
    {
        printf("%d ", i);
    }
    printf("\n");
}
```

/* Pattern 4: Reverse-number triangle */

```
void reverseTriangle(void)
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid number of rows!\n");
        return;
    }

    for (i = n; i >= 1; i--)
    {
```

```
    for (j = i; j >= 1; j--)  
    {  
        printf("%d ", j);  
    }  
    printf("\n");  
}  
}
```

4. Compile, Execute and Evaluate

```
(kali㉿kali)-[~/Desktop/decreasing_pattern]
$ vi decreasing_pattern.c

(kali㉿kali)-[~/Desktop/decreasing_pattern]
$ gcc decreasing_pattern.c -o decreasing_pattern

(kali㉿kali)-[~/Desktop/decreasing_pattern]
$ gcc -Wall -Wextra decreasing_pattern.c -o decreasing_pattern

(kali㉿kali)-[~/Desktop/decreasing_pattern]
$ ./decreasing_pattern
DECREASING PATTERNS
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 1
Enter the number of rows: 4
* * * *
* * *
* *
*

(kali㉿kali)-[~/Desktop/decreasing_pattern]
$ 2
2: command not found

(kali㉿kali)-[~/Desktop/decreasing_pattern]
$ ./decreasing_pattern
DECREASING PATTERNS
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter the number of rows: 4
1 2 3 4
1 2 3
1 2
1
```

```
1 2 3 4
1 2 3
1 2
1
```

```
(kali㉿kali)-[~/Desktop/decreasing_pattern]
```

```
$ ./decreasing_pattern
```

DECREASING PATTERNS

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle

Enter your choice: 3

Enter the number of rows: 4

```
4 4 4 4
3 3 3
2 2
1
```

```
(kali㉿kali)-[~/Desktop/decreasing_pattern]
```

```
$ ./decreasing_pattern
```

DECREASING PATTERNS

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle

Enter your choice: 4

Enter the number of rows: 4

```
4 3 2 1
3 2 1
2 1
1
```

```
(kali㉿kali)-[~/Desktop/decreasing_pattern]
```

```
$ ./decreasing_pattern
```

DECREASING PATTERNS

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern

```
4. Reverse-number triangle
Enter your choice: 1
Enter the number of rows: 1
*
```

```
(kali㉿kali)-[~/Desktop/decreasing_pattern]
$ ./decreasing_pattern
DECREASING PATTERNS
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 2
Enter the number of rows: 0
Invalid number of rows!
```

```
(kali㉿kali)-[~/Desktop/decreasing_pattern]
$ ./decreasing_pattern
DECREASING PATTERNS
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 3
Enter the number of rows: -2
Invalid number of rows!
```

```
(kali㉿kali)-[~/Desktop/decreasing_pattern]
$ ./decreasing_pattern
DECREASING PATTERNS
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
Enter your choice: 5
Invalid choice!
```

```
(kali㉿kali)-[~/Desktop/decreasing_pattern]
$
```